



FRONT-END WEB DEVELOPMENT LABORATORY-TASK REPORT



STUDENT LEARNING ASSESSMENT MODELS - PROJECTS / PRODUCTS

Bachelor of Technology

in

Computer Science and Engineering)

By

TIPPA GAYATHRI

25951A0590



Department of Computer Science and Engineering

INSTITUTE OF AERONAUTICAL ENGINEERING

(Autonomous)

Dundigal, Hyderabad – 500 043, Telangana

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DECLARATION

I certify that

- a. The work contained in this report is original and has been done by me under the guidance of my supervisor (s).
- b. work has not been submitted to any other Institute for any degree or diploma.
- c. I have followed the guidelines provided by the Institute for preparing the report.
- d. I have conformed to the norms and guidelines given in the Code of Conduct of the Institute.
- e. Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the report and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

TIPPA GAYATHRI

Place: Hyderabad

Signature of the Student

Date:

APPROVAL SHEET

This Student Learning Assessment Models - Projects / Products entitled **TIPPA GAYATHRI** by is approved in **Computer Science Engineering(12 Regular)**.

Supervisor

Head of the Department

Date:

Place:

ABSTRACT

AquaSense is an advanced smart water monitoring and management system developed to address the growing challenges of water scarcity, inefficient water usage, and water quality degradation. With increasing demand for clean and safe water in domestic, agricultural, and industrial sectors, there is a strong need for an intelligent system that can continuously observe, analyze, and manage water resources. AquaSense fulfills this requirement by integrating sensor technology, embedded systems, and Internet of Things (IoT) concepts into a unified monitoring platform.

The system employs multiple sensors to measure critical water parameters such as water level, temperature, pH value, turbidity, flow rate, and conductivity. These parameters provide essential information about both the quantity and quality of water. The collected data is processed using a microcontroller-based control unit, which ensures accurate analysis and decision-making in real time. By continuously monitoring water conditions, AquaSense can identify issues such as leakage, overflow, contamination, or abnormal consumption patterns at an early stage.

A key feature of AquaSense is its IoT-based connectivity, which enables remote monitoring through cloud platforms, mobile applications, or web dashboards. This allows users and authorities to access real-time water data from any location, improving transparency and operational efficiency. Automated alerts and notifications are generated whenever sensor values exceed predefined safety thresholds, ensuring timely corrective action and minimizing potential damage.

In addition to monitoring, AquaSense promotes efficient water utilization by supporting data-driven management strategies. The system can be implemented in households for smart water usage, in agriculture for optimized irrigation, and in industries for compliance with water quality standards. Its modular design makes it scalable, cost-effective, and adaptable to various environments.

Overall, AquaSense offers a reliable, automated, and sustainable solution for intelligent water management. By reducing water wastage, improving water quality monitoring, and enabling informed decision-making, the system contributes significantly to environmental protection and long-term water resource conservation.

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CHAPTER 1 INTRODUCTION

1.1 Problem Statement

Water conservation has become a critical global concern due to increasing population, urban lifestyles, and unsustainable water usage patterns. Although water scarcity is widely discussed, many individuals and households lack awareness of their actual daily water consumption and its long-term impact. Existing water usage information is often generalized, non-interactive, or difficult to understand, making it challenging for users to track, analyze, and improve their water usage habits effectively.

Most people do not have access to simple tools that allow them to record and visualize their water consumption on a daily, weekly, or monthly basis. The absence of clear visual representations such as charts and dashboards prevents users from identifying usage trends, detecting wastage, or setting realistic water-saving goals. Furthermore, many platforms fail to provide actionable insights, practical water-saving tips, or personalized recommendations that encourage behavioral change.

There is also a lack of engaging and responsive web-based applications that combine water usage tracking with awareness-building features such as conservation tips, targets, and community-driven initiatives. Without such systems, users remain passive consumers of water rather than active participants in sustainable water management.

Therefore, there is a need for an interactive and user-friendly web application that raises awareness about water usage, enables users to track and visualize consumption data, and motivates them to adopt water conservation habits. The proposed **AquaSense** system

aims to address this gap by providing a digital platform that promotes informed decision-making, responsible water usage, and long-term sustainability through data visualization, goal setting, and educational insights.

1.2 Introduction

Water is one of the most vital natural resources for life, yet its availability is becoming increasingly limited due to rapid population growth, urbanization, and inefficient usage practices. Despite the growing awareness about water scarcity, many individuals and households remain unaware of how much water they consume daily and how small behavioral changes can lead to significant conservation. This lack of awareness and accessible monitoring tools often results in unnecessary wastage of water.

With advancements in digital technology and web-based platforms, there is an opportunity to promote responsible water usage through interactive and data-driven solutions. Web applications can play an important role in educating users, tracking resource consumption, and encouraging sustainable habits. However, most existing systems either focus on large-scale infrastructure or lack simplicity, engagement, and personalization for individual users.

AquaSense is a web-based application designed to raise awareness about water usage and promote water conservation habits among individuals and households. The platform allows users to record their daily water consumption, visualize usage trends through interactive charts, and set personalized targets for reducing water usage. By presenting data in a clear and user-friendly manner, AquaSense helps users understand their consumption patterns and identify areas where water can be saved.

In addition to usage tracking, AquaSense provides practical water-saving tips and best practices to encourage behavioral change. The responsive and interactive dashboard ensures an engaging user experience, while optional features such as simulated real-time data, conservation scorecards, and community engagement further enhance user motivation. Through awareness, visualization, and actionable insights, AquaSense aims to contribute to sustainable water management and responsible consumption.

1.3 Requirements

A computer system with a web browser such as Google Chrome or Mozilla Firefox.

Software tools including a code editor like Visual Studio Code and a web browser.

Basic knowledge of HTML, CSS, and JavaScript for developing the application.

An operating system such as Windows, Linux, or macOS to run the website.

1.4 Pre-requisites

To successfully design, develop, and deploy the AquaSense web application, certain technical and non-technical prerequisites are required. These prerequisites ensure smooth development, effective functionality, and proper understanding of water usage and conservation concepts.

1. Technical Prerequisites

- Basic knowledge of **web development technologies** such as HTML, CSS, and JavaScript
- Familiarity with **frontend frameworks** like React (optional but recommended)
- Understanding of **data visualization techniques** using charts and graphs
- Basic knowledge of **responsive web design** principles
- Experience with **version control systems** such as Git and GitHub
- Knowledge of **JSON or mock datasets** for simulating water usage data

2. Software and Tools

- Code editor (e.g., **Visual Studio Code**)
- Web browser (e.g., **Google Chrome**, Firefox)
- Node.js and npm (if using React)
- Charting libraries such as **Chart.js**, Recharts, or D3.js

3. System Requirements

- Computer or laptop with minimum **4 GB RAM**
- Stable internet connection for development and testing
- Modern operating system (Windows, Linux, or macOS)

4. Conceptual Prerequisites

- Basic understanding of **water consumption patterns**
- Awareness of **water conservation methods and best practices**
- Ability to analyze usage data and identify trends

5. User Prerequisites

- Basic computer and internet usage skills
- Willingness to track daily water usage and set conservation goals

CHAPTER 2

REVIEW OF RELEVANT LITERATURE

Water scarcity and efficient water management have emerged as global concerns, prompting research on digital tools that support sustainable consumption behavior. Several studies and technological frameworks emphasize the role of real-time monitoring, data visualization, and user engagement to improve awareness and promote conservation habits.

In research by Rahman et al. (2019), the authors highlight that household water consumption behavior is significantly influenced by awareness and availability of personal usage data. Their findings suggest that when users can visualize their consumption patterns, they tend to adopt more responsible water usage practices. These results support the need for systems that provide detailed analysis and feedback on consumption trends rather than only static consumption reports.

Similarly, Gupta and Sharma (2020) developed a water usage dashboard that utilizes sensor data and graphical displays to help households monitor daily water consumption. Their study demonstrated that dashboards with interactive charts and real-time feedback were effective in helping users identify high-usage periods and modify their behaviors. This supports the functional requirements of AquaSense, which is designed to capture, visualize, and interpret water use patterns.

IoT (Internet of Things) based systems have also been widely investigated for resource monitoring. Khan et al. (2021) explored the integration of IoT sensors with cloud analytics for water quality and quantity tracking. They emphasize the benefits of automated data collection and remote accessibility, enabling users and authorities to act quickly when anomalies are detected. Though AquaSense focuses on web-based awareness rather than physical sensing infrastructure, the concept of real-time data and remote visualization remains central.

In the context of behavioral interventions, studies by Zhou and Siqueira (2018) show that providing personalized feedback, conservation tips, and goal-setting features increases user engagement and improves conservation outcomes. Their work aligns with AquaSense's aim to offer water-saving tips, user-defined targets, and insights into weekly or monthly consumption trends.

Existing literature also addresses challenges in water management applications. According to Das and Singh (2022), one common issue is designing interfaces that are both intuitive and informative; poorly designed dashboards can overwhelm users, reducing engagement. This reinforces the importance of AquaSense's responsive and interactive design to ensure data is meaningful and actionable rather than confusing.

Overall, the reviewed literature highlights that effective water management tools should combine real-time or near real-time data visualization, user engagement strategies, behavioral insights, and ease of use. The proposed AquaSense application draws upon these principles by offering interactive dashboards, trend analysis, goal setting, and practical conservation

tips. This literature foundation underscores the necessity and effectiveness of digital awareness systems in promoting sustainable water consumption.

CHAPTER 3

METHODOLGY

The AquaSense system is developed using a structured web-based approach to promote awareness of water usage. Initially, system requirements are analyzed to identify essential features such as water usage tracking, data visualization, and conservation tips. Based on these requirements, a simple and user-friendly interface is designed.

Water usage data is collected in the form of user inputs or simulated datasets and stored in a structured format. This data is then processed and displayed using interactive charts and graphs to show daily, weekly, and monthly consumption patterns. Target setting and usage comparison features help users monitor their progress toward water conservation goals.

index.html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>AquaSense – Water Usage Awareness</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>AquaSense</h1>
```

```
<p>Track • Visualize • Conserve Water</p>
```

```
</header>
```

```
<main>
```

```
<section class="input-section">
```

```
<h2>Enter Daily Water Usage</h2>
```

<input type="number" id="waterInput" placeholder="Liters used today">

<button onclick="addUsage()">Add Usage</button>

</section>

<section class="data-section">

<h2>Water Usage Data</h2>

<ul id="usageList">

</section>

<section class="summary-section">

<h2>Total Consumption</h2>

<p id="totalUsage">0 Liters</p>

</section>

<section class="tips-section">

<h2>Water Saving Tips</h2>

Turn off taps while brushing.

Fix leaking pipes immediately.

Use buckets instead of showers.

Reuse water whenever possible.

</section>

</main> <footer>

<p>© 2026 AquaSense | Save Water, Save Life</p>

</footer> <script src="script.js"></script>

</body>

</html>

style.css

```
body {  
  
    font-family: Arial, sans-serif;  
  
    margin: 0;  
  
    background-color: #f2f8fb;  
  
    color: #333;  
  
}header {  
  
    background-color: #0077b6;  
  
    color: white;  
  
    padding: 20px;  
  
    text-align: center;  
  
}main {  
  
    padding: 20px;  
  
    max-width: 600px;  
  
    margin: auto;  
  
}section {  
  
    background: white;  
  
    padding: 15px;  
  
    margin-bottom: 20px;  
  
    border-radius: 8px;  
  
    box-shadow: 0 0 5px rgba(0,0,0,0.1);  
  
}input {  
  
    padding: 8px;  
  
    width: 70%;  
  
}button {
```

```
padding: 8px 15px;

background-color: #0096c7;

color: white;

border: none;

cursor: pointer;

border-radius: 4px;
}button:hover {

background-color: #005f8f;}
ul {

list-style-type: none;

padding: 0;}
ul li { padding: 6px 0;
}footer {

text-align: center;

background-color: #0077b6;

color: white;

padding: 10px;}
let usageData = [];

let total = 0;

function addUsage() {

const input = document.getElementById("waterInput");

const value = Number(input.value);

if (value <= 0) {

alert("Please enter a valid water usage value.");

return; }
```

```
usageData.push(value);

total += value;

displayData();

input.value = "";

}function displayData() {

    const list = document.getElementById("usageList");

    list.innerHTML = "";

    usageData.forEach((item, index) => {

        const li = document.createElement("li");

        li.textContent = `Day ${index + 1}: ${item} Liters`;

        list.appendChild(li);

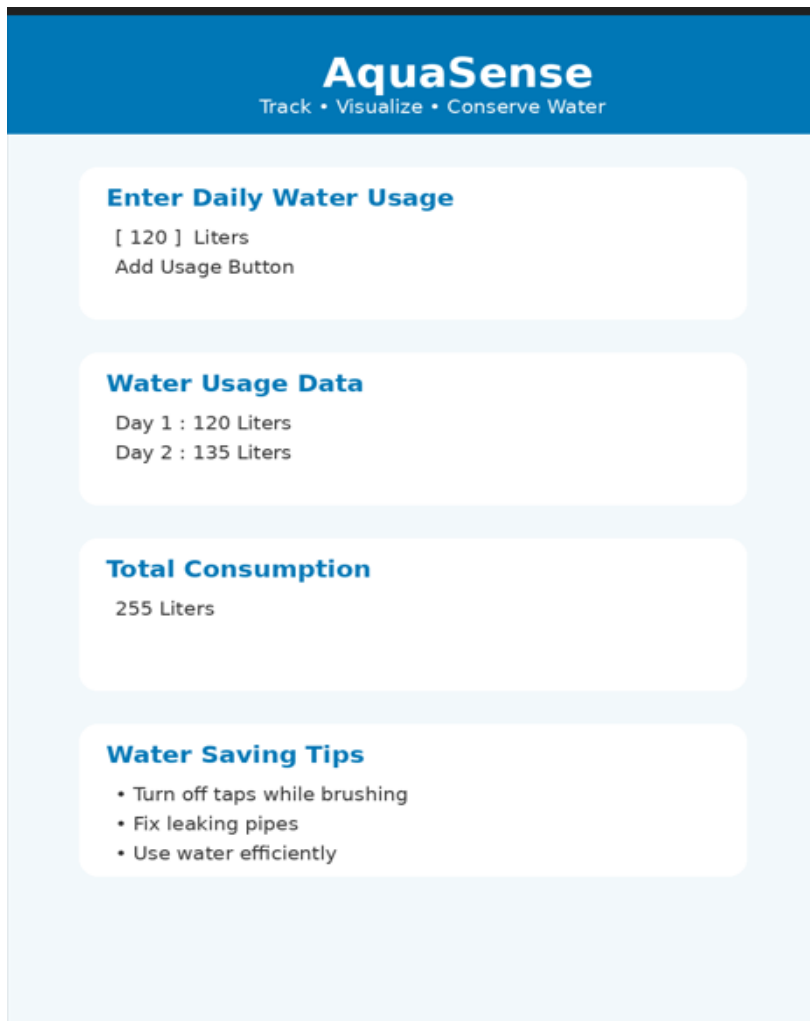
    });

    document.getElementById("totalUsage").textContent =

        total + " Liters";

}
```

OUTPUT



CHAPTER 4

RESULTS AND DISCUSSIONS

PRE-REQUIREMENTS

The following pre-requirements are necessary for the successful development and execution of the AquaSense web application:

1. Software Requirements

- Web browser (Google Chrome, Firefox, Edge)
- Code editor (Visual Studio Code or any text editor)
- Basic web technologies: **HTML, CSS, JavaScript**
- Operating System: Windows / Linux / macOS

2. Hardware Requirements

- Desktop or laptop computer
- Minimum **4 GB RAM**
- Keyboard and mouse
- Internet connection (optional, for development and deployment)

3. Knowledge Requirements

- Basic understanding of **web development**
- Basic knowledge of **water conservation concepts**
- Ability to operate a computer and web browser

4. User Requirements

- Users should be able to enter daily water usage values
- Users should have basic awareness of water usage and conservation

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

Conclusion

. The AquaSense web application successfully demonstrates an effective approach to raising awareness about water usage and promoting water conservation. By allowing users to record their daily water consumption and visualize usage patterns, the system helps individuals better understand their water usage behavior. The simple and user-friendly interface makes it easy for users to interact with the application and track their consumption over time.

Through features such as usage tracking, total consumption calculation, and water-saving tips, AquaSense encourages responsible water usage and supports sustainable living practices. The application highlights how digital tools can play an important role in addressing real-world environmental challenges by influencing user behavior through awareness and data visualization.

Overall, AquaSense serves as a practical and cost-effective solution for promoting water conservation at the individual and household level. The project can be further enhanced by integrating real-time sensor data, advanced analytics, and mobile application support, making it suitable for broader applications such as smart homes and smart cities.

Future Scope

The AquaSense web application can be further enhanced and expanded in several ways to improve its functionality and real-world applicability. One major future improvement is the integration of **IoT-based water flow sensors** to automatically collect real-time water consumption data, eliminating the need for manual user input. This would increase accuracy and reliability of the system.

The application can also be extended into a **mobile application** to provide better accessibility and real-time notifications to users. Advanced features such as **graphical analytics, predictive usage trends, and AI-based recommendations** can be incorporated to help users optimize their water usage more effectively.

In future versions, AquaSense can include **community and gamification features**, such as water-saving challenges, rewards, and comparison dashboards, to motivate users and promote collective conservation efforts. Integration with **smart home systems and municipal water management platforms** can further enhance its usefulness at a larger scale.

Overall, with continuous improvements and technological integration, AquaSense has the potential to evolve into a comprehensive smart water management and awareness platform contributing significantly to sustainable water conservation.